NETWORK DEVICES

1.Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

ANS:

Simple Home Network Diagram

Internet

Router

Connects the home network to the Internet

Network Switch

Provides additional Ethernet ports

Laptop

Ethernet cable

Smartphone

Wi-Fi via router

Smart TV

Ethernet cable

Explanation:

1. Router: Connects the home network to the Internet and provides Wi-Fi.
2. Switch: Connects multiple wired devices to the same network.
3. Laptop: Connects to the switch using an Ethernet cable.
4. Smartphone: Connects wirelessly to the router through Wi-Fi.
5. Smart TV: Connects to the switch using an Ethernet cable.

2.Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

ANS:

Comparison Table of Network Devices

Network Device	**Main Function**	**Real-World Use Case**

Router	Connects different networks and provides Internet access.	Using Wi-Fi to watch videos on YouTube or Netflix.
Switch	Connects multiple devices within a LAN.	Connecting gaming PCs in a gaming café for multiplayer LAN gaming.
Hub	Connects multiple devices and broadcasts data to all connected ports.	Connecting older computers in a small computer lab for basic network communication.
Access Point (AP)	Provides wireless network access to devices.	Connecting smartphones and laptops to college Wi-Fi to use WhatsApp or Google Classroom.

3.Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

ANS:

Scenario: Ordering Food on Zomato

When I order food using the Zomato app on my smartphone, my phone sends the order data through Wi-Fi to the access point, which connects my phone to the local network. The switch forwards the data between devices within the local network, if needed. The router forwards the data toward the Internet through my Internet Service Provider (ISP). Zomato's servers receive the request and process my order. The confirmation then travels back through the Internet and router to my phone.

**4.Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.**

ANS:

Two Network Devices to Connect a Gaming PC to the Internet

1. Powerline Adapter

- It uses the existing electrical wiring in the house to transmit network data.
- Connect one adapter to the router using an Ethernet cable and plug it into a power socket.
- Plug the second adapter near the gaming PC and connect it using an Ethernet cable.

- Use: Provides a wired connection for online gaming without running a long Ethernet cable.

2. Wi-Fi Extender with an Ethernet Port

- It extends the router's Wi-Fi coverage to areas farther away.
- Place the extender between the router and the gaming PC, where it receives a good Wi-Fi signal.
- Connect the gaming PC's Ethernet port to the extender using an Ethernet cable.
- Use: Allows the gaming PC to access the Internet without a direct cable to the router.